OPERATING SYSTEM LAB ASSIGNMENT 1

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
(base) rvu@rvu-OptiPlex-SFF-7020:~$ cd wruthambara
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ nano ruthu.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ bash ruthu.sh
Parent PID:4435
Child Process Created
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps
    PID TTY          TIME CMD
   3284 pts/0    00:00:00 bash
   4459 pts/0    00:00:00 ps
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
rvu         3284    3277  0 19:35 pts/0    00:00:00 bash
rvu         4472    3284  0 19:48 pts/0    00:00:00 ps -f
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 19:35 ?        00:00:01 /sbin/init splash
root           2       0  0 19:35 ?        00:00:00 [kthreadd]
root           3       2  0 19:35 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 19:35 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 19:35 ?        00:00:00 [kworker/R-sync_wq]
```

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
  GNU nano 7.2                                                    ruthu.sh
#!/bin/bash
echo "Parent PID:$$"
sleep 5 &
echo "Child Process Created"
```

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
root        4140       2  0 19:43 ?        00:00:00 [kworker/4:0-i915-unordered]
root        4439       2  0 19:46 ?        00:00:00 [kworker/10:0-mm_percpu_wq]
rvu         4482    3284  0 19:48 pts/0    00:00:00 ps -ef
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ top

top - 20:01:07 up 25 min,  1 user,  load average: 0.22, 0.11, 0.09
Tasks: 366 total,   1 running, 364 sleeping,   0 stopped,   1 zombie
%Cpu(s):  0.4 us,  0.4 sy,  0.0 ni, 99.2 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15677.2 total,  13174.7 free,   1746.1 used,   1316.5 buff/cache
MiB Swap:   4096.0 total,   4096.0 free,      0.0 used.  13931.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2449 rvu       20   0 4959816 346304 139160 S   4.0   2.2   0:32.19 gnome-shell
   2972 rvu       20   0  533016  15012   7772 S   4.0   0.1   0:47.54 speech-dispatch
   2181 rvu        9 -11  135780  25120  10084 S   2.0   0.2   0:28.73 pipewire-pulse
   2175 rvu        9 -11  119072  17620   9352 S   1.7   0.1   0:22.06 pipewire
   3277 rvu       20   0  705964  64304  50172 S   1.7   0.4   0:09.23 gnome-terminal-
   2719 rvu       20   0  762908  69912  35072 S   0.7   0.4   0:15.26 orca
    191 root       0 -20       0      0      0 I   0.3   0.0   0:01.52 kworker/u81:0-i915_flip
    707 root      20   0       0      0      0 I   0.3   0.0   0:00.28 kworker/u80:6-flush-259:0
      1 root      20   0   23952  15216   9512 S   0.0   0.1   0:01.97 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.16 kworker/0:1-events
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.02 ksoftirqd/0
     18 root      20   0       0      0      0 I   0.0   0.0   0:00.81 rcu_preempt
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/1
```

rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara

```
  2449 rvu       20   0 4959816 346304 139160 S   4.0   2.2   0:32.19 gnome-shell
  2972 rvu       20   0  533016  15012   7772 S   4.0   0.1   0:47.54 speech-dispatch
  2181 rvu        9 -11  135780  25120  10084 S   2.0   0.2   0:28.73 pipewire-pulse
  2175 rvu        9 -11  119072  17620   9352 S   1.7   0.1   0:22.06 pipewire
  3277 rvu       20   0  705964  64304  50172 S   1.7   0.4   0:09.23 gnome-terminal-
  2719 rvu       20   0  762908  69912  35072 S   0.7   0.4   0:15.26 orca
   191 root       0 -20       0      0      0 I   0.3   0.0   0:01.52 kworker/u81:0-i915_flip
   707 root      20   0       0      0      0 I   0.3   0.0   0:00.28 kworker/u80:6-flush-259:0
     1 root      20   0   23952  15216   9512 S   0.0   0.1   0:01.97 systemd
     2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
     3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
     4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
     5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
     6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
     7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
     8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
    10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    11 root      20   0       0      0      0 I   0.0   0.0   0:00.16 kworker/0:1-events
    13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
    14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
    15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
    16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
    17 root      20   0       0      0      0 S   0.0   0.0   0:00.02 ksoftirqd/0
    18 root      20   0       0      0      0 I   0.0   0.0   0:00.81 rcu_preempt
    19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/1
    20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
    21 root      rt   0       0      0      0 S   0.0   0.0   0:00.01 migration/0
    22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
    23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
    24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ jobs
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ bg % 1
bash: bg: %: no such job
bash: bg: 1: no such job
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ fg % 1
bash: fg: %: no such job
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$
```

Tue Feb 17 20:43:26

rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara

```
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ echo $$
4975
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ sleep 30
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ sleep 60 &
[1] 5249
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ jobs
[1]+  Running                 sleep 60 &
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
rvu         4975    4967  0 20:32 pts/0    00:00:00 bash
rvu         5294    4975 33 20:43 pts/0    00:00:00 ps -f
[1]+  Done                    sleep 60
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 20:30 ?        00:00:01 /sbin/init splash
root           2       0  0 20:30 ?        00:00:00 [kthreadd]
root           3       2  0 20:30 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 20:30 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 20:30 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 20:30 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 20:30 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 20:30 ?        00:00:00 [kworker/R-netns]
root          10       2  0 20:30 ?        00:00:00 [kworker/0:0H-events_highpri]
root          11       2  0 20:30 ?        00:00:00 [kworker/0:1-events]
root          13       2  0 20:30 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 20:30 ?        00:00:00 [rcu_tasks_kthread]
root          15       2  0 20:30 ?        00:00:00 [rcu_tasks_rude_kthread]
root          16       2  0 20:30 ?        00:00:00 [rcu_tasks_trace_kthread]
root          17       2  0 20:30 ?        00:00:00 [ksoftirqd/0]
root          18       2  0 20:30 ?        00:00:00 [rcu_preempt]
root          19       2  0 20:30 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/1]
root          20       2  0 20:30 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          21       2  0 20:30 ?        00:00:00 [migration/0]
root          22       2  0 20:30 ?        00:00:00 [idle_inject/0]
root          23       2  0 20:30 ?        00:00:00 [cpuhp/0]
root          24       2  0 20:30 ?        00:00:00 [cpuhp/2]
root          25       2  0 20:30 ?        00:00:00 [idle_inject/2]
```

```
Tue Feb 17  20:44:29
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
rvu         4702    2384  0 20:31 ?        00:00:00 /usr/bin/update-notifier
rvu         4763    3642  0 20:31 ?        00:00:00 /snap/firefox/7423/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:44322 -prefM
rvu         4841    2368  0 20:31 ?        00:00:00 /usr/libexec/gvfsd-network --spawner :1.20 /org/gtk/gvfs/exec_spaw/1
rvu         4851    2368  0 20:31 ?        00:00:00 /usr/libexec/gvfsd-dnssd --spawner :1.20 /org/gtk/gvfs/exec_spaw/2
rvu         4967    2140  0 20:32 ?        00:00:01 /usr/libexec/gnome-terminal-server
rvu         4975    4967  0 20:32 pts/0    00:00:00 bash
root        5185       2  0 20:36 ?        00:00:00 [kworker/6:0]
root        5259       2  0 20:40 ?        00:00:00 [kworker/4:0-cgroup_destroy]
root        5260       2  0 20:40 ?        00:00:00 [kworker/4:3-events]
rvu         5304    4975  0 20:43 pts/0    00:00:00 ps -ef
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -ef --forest
UID          PID    PPID  C STIME TTY          TIME CMD
root           2       0  0 20:30 ?        00:00:00 [kthreadd]
root           3       2  0 20:30 ?        00:00:00  \_ [pool_workqueue_release]
root           4       2  0 20:30 ?        00:00:00  \_ [kworker/R-rcu_gp]
root           5       2  0 20:30 ?        00:00:00  \_ [kworker/R-sync_wq]
root           6       2  0 20:30 ?        00:00:00  \_ [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 20:30 ?        00:00:00  \_ [kworker/R-slub_flushwq]
root           8       2  0 20:30 ?        00:00:00  \_ [kworker/R-netns]
root          10       2  0 20:30 ?        00:00:00  \_ [kworker/0:0H-events_highpri]
root          11       2  0 20:30 ?        00:00:00  \_ [kworker/0:1-events]
root          13       2  0 20:30 ?        00:00:00  \_ [kworker/R-mm_percpu_wq]
root          14       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_kthread]
root          15       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_rude_kthread]
root          16       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_trace_kthread]
root          17       2  0 20:30 ?        00:00:00  \_ [ksoftirqd/0]
root          18       2  0 20:30 ?        00:00:00  \_ [rcu_preempt]
root          19       2  0 20:30 ?        00:00:00  \_ [rcu_exp_par_gp_kthread_worker/1]
root          20       2  0 20:30 ?        00:00:00  \_ [rcu_exp_gp_kthread_worker]
root          21       2  0 20:30 ?        00:00:00  \_ [migration/0]
root          22       2  0 20:30 ?        00:00:00  \_ [idle_inject/0]
root          23       2  0 20:30 ?        00:00:00  \_ [cpuhp/0]
root          24       2  0 20:30 ?        00:00:00  \_ [cpuhp/2]
root          25       2  0 20:30 ?        00:00:00  \_ [idle_inject/2]
root          26       2  0 20:30 ?        00:00:00  \_ [migration/2]
root          27       2  0 20:30 ?        00:00:00  \_ [ksoftirqd/2]
root          28       2  0 20:30 ?        00:00:00  \_ [kworker/2:0-i915-unordered]
```

```
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ nano child_demo.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ./child_demo.sh
bash: ./child_demo.sh: Permission denied
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ chmod +x child_demo.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ./child_demo.sh
Parent PID:5821
UID          PID    PPID  C STIME TTY          TIME CMD
rvu         4975    4967  0 20:32 pts/0    00:00:00 bash
rvu         5821    4975  0 20:56 pts/0    00:00:00  \_ /bin/bash ./child_demo.sh
rvu         5822    5821  0 20:56 pts/0    00:00:00      \_ sleep 200
rvu         5823    5821  0 20:56 pts/0    00:00:00      \_ sleep 250
rvu         5824    5821  0 20:56 pts/0    00:00:00      \_ ps -f --forest
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ sleep 100 &
[1] 5844
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ sleep 120 &
[2] 5859
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ sleep 140 &
[3] 5873
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -f --forest
UID          PID    PPID  C STIME TTY          TIME CMD
rvu         4975    4967  0 20:32 pts/0    00:00:00 bash
rvu         5844    4975  0 20:56 pts/0    00:00:00  \_ sleep 100
rvu         5859    4975  0 20:56 pts/0    00:00:00  \_ sleep 120
rvu         5873    4975  0 20:56 pts/0    00:00:00  \_ sleep 140
rvu         5895    4975 99 20:56 pts/0    00:00:00  \_ ps -f --forest
rvu         5823    2140  0 20:56 pts/0    00:00:00 sleep 250
rvu         5822    2140  0 20:56 pts/0    00:00:00 sleep 200
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$
```

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
  GNU nano 7.2                                child_demo.sh
#!/bin/bash
echo "Parent PID:$$"
sleep 200 &
sleep 250 &
ps -f --forest
```

nano filename.sh -> Creates Shell Script
Bash -> Parent process
cd -> change directories
ps -> Display process Status & if is a child process
pstree -> Parent child relationship
top -> Shows Table of Process
job -> list of background processes
fg %1 -> Brings job to foreground
bg %1 -> Resumes job in background
Kill -> Terminates process
Sleep 30: Pause processes for 30 Seconds
Sleep 60 &: Run the process in background
ps -f: Full format listing
ps -ef: Shows every process on the System
ps -ef --forest: list of all running process in hierarchical tree
echo $$: Displays process ID (PID)
Sleep 100 &: Sleeps / pause for 100 Seconds
ps -f --forest: Display current Running file